The Great Repricing

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Abstract

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1 Important Paper and Random Thoughts

Factors/anamolies have had a decline in performance. This exact date is roughly around decimalization and the years 2000s.

- Chordia et al. (2014) try to attribute it all to arbitrage activity. They take short interest, hedge fund AUM and turnover as proxy for arb, then use a VAR setup to conclude pseudo causality. They and many others mostly argue that it is the decline comes from the α part of the pricing equation. They argue it can't all be systematic, and that there aren't good explanations. But their SDF sucked and their coefficients were static. **Analysis is not good on many levels. The study takes realized returns as a proxy for expected returns, uses innovations to proxy for exogenous shocks. Completely ignores the implication of the systematic decline of RP.**
- **Penase and Binsbergen, also Pénasse (2022)** The decay of α (same argument applies to the mean of asset returns) pushes up prices up temporarily (FOA duration times change) when it starts to decline (change could have happened earlier). In the follow-up paper, Penase has a test given declining alphas to see if anomalies are dead, he finds that a lot of the same survive.

Directions:

- Do the change points differ if you adjust for some costs?
- Does the change still come from alpha if adjusting for a pricing model with time-varying betas and high SR SDF? (In-sample PCA, shows that it is in fact the common component that experienced the change)
- Extend of decay as a function of liquidity constraints (turnover, factors without TC, overal trading cost measure)
- Important to note we are different than Penase as we care about when the means/performance changes, not the death of anomalies. In fact, I would argue that most are alive and kicking.
- Can we derive a test of change, rather than factor death (consistent with non-stationarity argument of Pénasse (2022)), to show the repricing in 2000.
- Build a learning based model from returns and chars (similar to Marrow and Nagel (2024)). Can investors figure out the repricing in real time? Online change point detection exercise rather than a t-stat/comparison with zero. If so, then it would make sense for the savviest of investors to change their positions and accentuate the decline.
- The low-interest-rate environment could have also played a role in the decline of anoma-

lies by enabling arbitrage. Shocks to the short-rate vs long-rate of interest rates could help identify a potential tool. Using good proxies for expected returns and investor learning, both Marrow and Nagel (2024) maybe even **?**

- Ineasltic markets
- Looking at flows: Chordia et al. (2014) says hedge fund AUM went up in this period, but who exactly bought what? Are their any breaks in their flows and char exposures? Who are the key players who started buying?
- Could decimalization be a convenient scapegoat? Both TC and alpha decay bias results to the left side (saying the break happens earlier). Why as it could be later. Maybe even around the GFC. The role of intermedieres
- changes in demands from holdings
- should their be a jump in the point?
- cost of return goes down, returns of makret maker goes down, because they gain the costs.
- market maker proxy portfolio
- ipca with tc
- ipca with trend

2 Our Potential Contributions

2.1 Systematic not Idiosyncratic

It's true that all the factors have declined, but at the same time, we know factors move together a lot and have strong common components (Kozak et al., 2020). Hence, it isn't clear whether or not this join component or the residuals are what moved. Indeed, our empirical evidence would suggest that most SDFs exhibit a change at the same time. However, residuals from a static regression still exhibit changes. I conjecture that this is mostly driven by time-varying betas and can be addressed with some form of some kind of IPCA with shrinkage (for example Bybee et al. (2023)). Additionally, one could learn the betas with characteristics using a single-layer neural network (for example Gu et al. (2021)). I define characteristics at the factor level as the weighted sum of stock characteristics. The literature has shown that factors as also predictable and can be timed with chars so this approach could make sense (for example Haddad et al. (2020)). We can then show

- Factors are priced (indistinguishable from zero) by the SDF (similar to the results in Kelly et al. (2019))
- Furthermore, change points become much less dispersed around the middle and none can be detected.

Thoughts: Just because something is systematic does not mean it is related to risk; this for sure seems more like mispricing. Somehow related to the fundamental nature of how inefficient the market can be. Cross sectional in-efficiency. People could make money on this high sharpe strategy then they overinvested and it want down it's the same story doesn't change if it's systematic.

2.2 Is it about transaction costs?

- Structure model of SDF with costs. Calibrate to real data, how much of the decline can be explained by adhoc cost measures. Must also account for potential change in demand as well.
- It's not just about the costs, build assets either with turnover, size, and illiquidity measures, show the break is still there.
- Build factors with and without costs
- See also Federico's cost aware factors
- costs can be viewed as the market maker's portfolio
- can we build an IPCA with costs

2.3 The role of demand

At the heart of all this discussion lies investor flows and quantities. Someone had to have bought the chars.

- Can we get a mapping from holdings to chars? Was there a break point somewhere? Who decided to buy more?
- Need to think about high-frequency players and their ability to quickly offload (imputation with asset returns?) Especially important after 2008
- IPCA kinda setup for demand and chars (Read the Koijen and Yogo carefully)
- Ask Yinan, he seems to have worked with quantities before (read his paper)
- How does alpha decay map into demand?

- There was an increase in factor returns in line with the alpha decay story starting in the 1990s. People started buying this stuff more aggressively, pushing realized returns up and depressing future returns.

2.4 Real Time Inference and statistical testing

To make any causal statements, we need two things:

- A measure of expected returns (take Kelly and Pruitt (2013), also real time learning like Marrow and Nagel (2024) could be interesting)
- Some form of exogenous shocks (candidates FOMC, short vs long rate, policy changes, demand instruments)

For testing, we need to take the considerations of Pénasse (2022) into account.

- Build an IPCA with trend or break point, build tests with bootstrap
- Build a test like Marrow and Nagel (2024) but ulitimately change the reference point to cumsum
- Online change point detection, can it help with SDF performance? Do breakpoints help escape potentially risky situations?

2.5 Economic Drivers

- The switch from Bank intermediation to hedge funds could be important...
- Could have started earlier (momentum paper was written)
- Glass–Steagall Act
- TC

2.6 Streaky returns and the role of break points

Proposition 1 (Single mean break inflates the variance ratio)
Let $\{r_t\}_{t=1}^{T}$ be returns with a single deterministic mean break at $\tau \in \{1, \ldots, T-1\}$:

$$r_t = \mu_t + \varepsilon_t, \qquad \mu_t = \begin{cases} \mu_1, & t \le \tau, \\ \mu_2, & t > \tau, \end{cases}$$

where $\{\varepsilon_t\}$ are i.i.d. with $\mathbb{E}[\varepsilon_t] = 0$ and $\mathrm{Var}(\varepsilon_t) = \sigma^2 \in (0, \infty)$. Let $\Delta := \mu_2 - \mu_1 \neq 0$ and $p := \tau/T \in (0, 1)$. For a fixed horizon $q \geq 2$, define the variance ratio

$$VR_T(q) := \frac{\mathrm{Var}_t\Big(\sum_{j=0}^{q-1} r_{t+j} \Big)}{q\,\mathrm{Var}_t(r_t)},$$

where $\mathrm{Var}_t(\cdot)$ is the empirical variance across admissible t.

Then as $T \to \infty$ with q fixed,

$$VR_T(q) \;\longrightarrow\; 1 + \frac{(q-1)\,\Delta^2\,p(1-p)}{\sigma^2 + \Delta^2\,p(1-p)} \;>\; 1.$$

Sketch of proof. **Step 1 (Denominator).** Since $r_t = \mu_t + \varepsilon_t$, we have

$$\mathrm{Var}_t(r_t) = \mathrm{Var}_t(\mu_t) + \mathrm{Var}(\varepsilon_t).$$

Over the whole sample, $\mu_t = \mu_1$ for a fraction p of dates and μ_2 for $1 - p$. Thus

$$\mathrm{Var}_t(\mu_t) = p(1-p)(\mu_2 - \mu_1)^2 = \Delta^2 p(1-p).$$

Therefore

$$\mathrm{Var}_t(r_t) = \sigma^2 + \Delta^2 p(1-p).$$

Step 2 (Numerator). Write the q-period sum as

$$S_t(q) = \sum_{j=0}^{q-1} r_{t+j} = M_t(q) + E_t(q), \qquad M_t(q) = \sum_{j=0}^{q-1} \mu_{t+j}, \quad E_t(q) = \sum_{j=0}^{q-1} \varepsilon_{t+j}.$$

By independence, $\mathrm{Var}(E_t(q)) = q\sigma^2$ and $\mathrm{Cov}(M_t(q), E_t(q)) = 0$.

For all but $q-1$ "straddling" windows around τ, the block mean is constant: $M_t(q) = q\mu_1$ (pre-break) or $M_t(q) = q\mu_2$ (post-break). Hence, across t, the variance of $M_t(q)$ equals

$$\mathrm{Var}_t(M_t(q)) = q^2 \Delta^2 p(1-p) + O\big(\tfrac{q}{T}\big).$$

Step 3 (Assemble ratio). Therefore

$$\mathrm{Var}_t(S_t(q)) = q\sigma^2 + q^2 \Delta^2 p(1-p) + O\big(\tfrac{q}{T}\big).$$

Dividing by $q\,\mathrm{Var}_t(r_t)$ gives

$$VR_T(q) = \frac{\sigma^2 + q\Delta^2 p(1-p)}{\sigma^2 + \Delta^2 p(1-p)} + O\big(\tfrac{q}{T}\big) = 1 + \frac{(q-1)\Delta^2 p(1-p)}{\sigma^2 + \Delta^2 p(1-p)} + O\big(\tfrac{q}{T}\big)\,.$$

As $T \to \infty$, the $O(q/T)$ term vanishes and the limit is strictly greater than 1 whenever $\Delta \neq 0$. $\square$

Intuition. The denominator $\mathrm{Var}(r_t) = \sigma^2 + \Delta^2 p(1-p)$ sees the mean-difference once. The numerator, $\mathrm{Var}(S_t(q))$, sees it q times inside each block, giving $q^2\Delta^2 p(1-p)$. After dividing by q, a leftover factor of $(q-1)$ remains, so $VR(q) > 1$. Straddling windows are only $O(q/T)$ of the sample and vanish asymptotically.

2.6.1 Streaky Returns in JKP Factors

Variance Ratio ($q = 12$) no Overlap, vs Sharpe Ratio of JKP Factors.

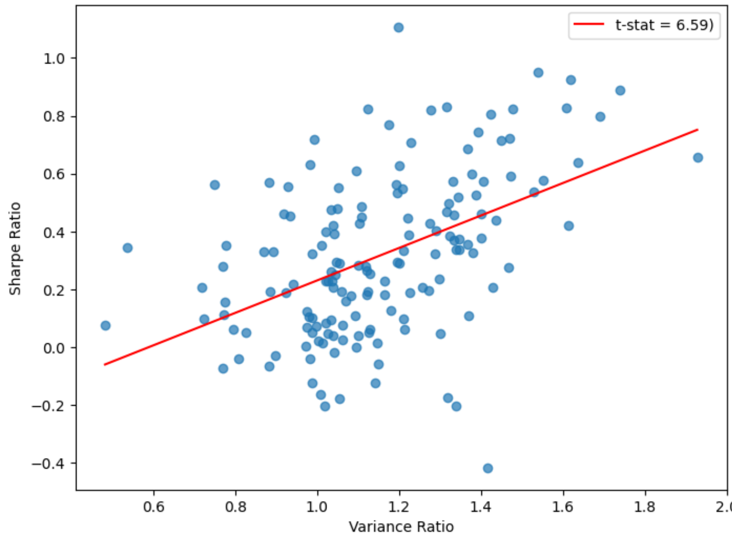

Figure 1: Full Sample

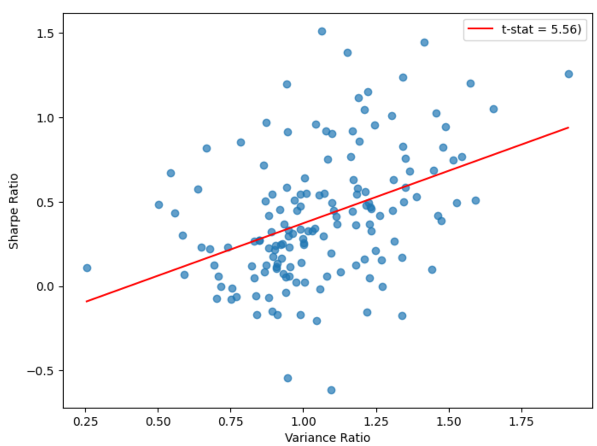

Figure 2: Before 2002

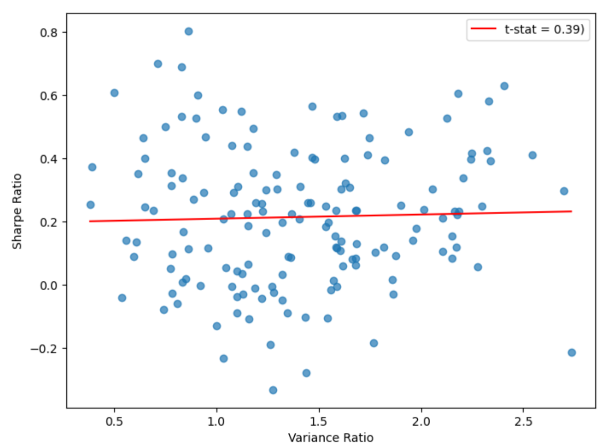

Figure 3: After 2002

Variance Ratio ($q = 12$) no Overlap, vs Sharpe Ratio of JKP Factors Vol Managed.

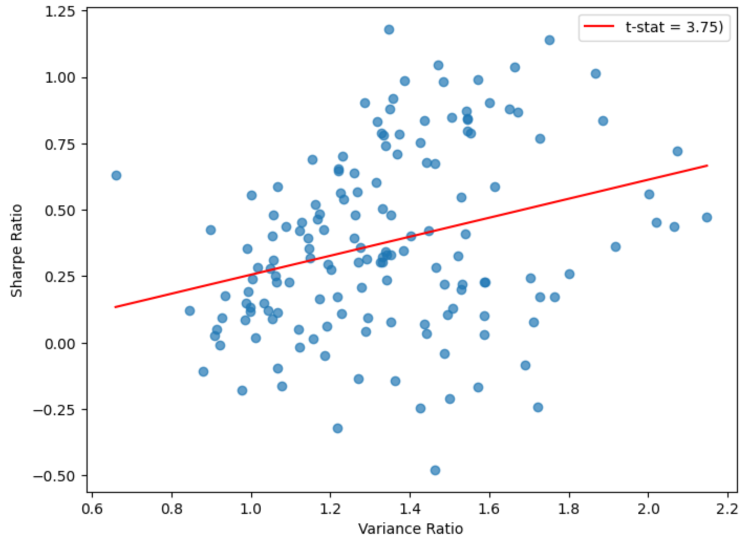

Figure 4: Full Sample

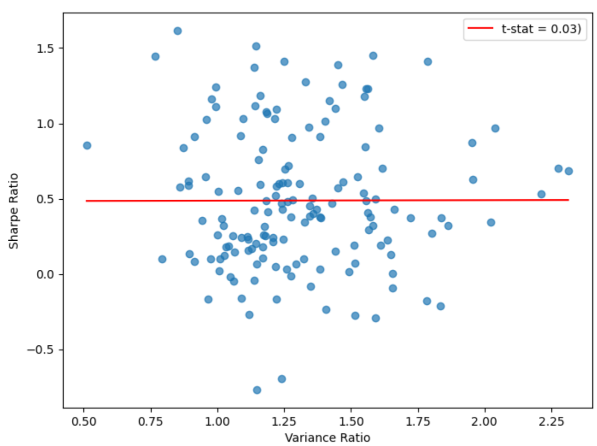

Figure 5: Before 2002

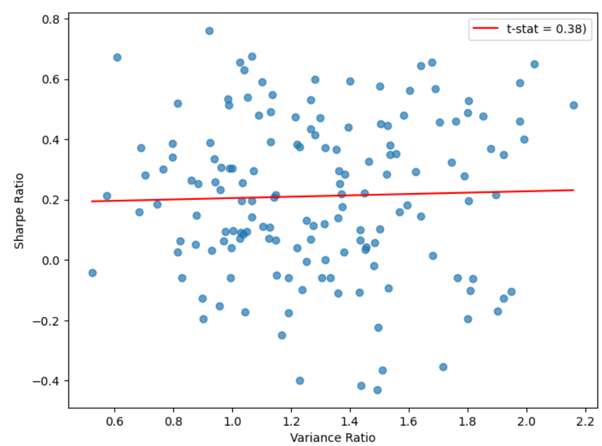

Figure 6: After 2002

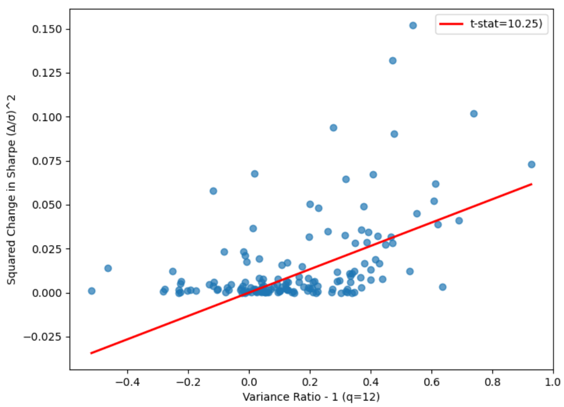

Figure 7: Changes in Sharpe ratio (pre-post break) vs variance ratio

2.7 A Simple Test of Break Point

Before, I relied on change-point detection methods such as the CUSUM test, which search over all possible break dates in the time series to identify the most likely point of change. Because such tests must consider every potential breakpoint, they are generally weak in finite samples and rarely lead to strong rejections. The problem is compounded when testing across all 153 factors and many candidate SDFs, since this raises severe multiple-testing concerns.

Given that we already have strong prior evidence on the timing of the change (around 2001), and in order to exploit the joint power of the full cross section, a more natural approach is to use a simple pooled regression with a post-break dummy. This framework is familiar to most economists, and it conveniently allows for clustering at the time level as well as the inclusion of factor fixed effects to absorb differences in pre-break risk premiums:

Definition 1 (Simple pooled breakpoint test)**.** Let $r_{i,t}$ denote the excess return of factor $i = 1, \ldots, N$ in month $t = 1, \ldots, T$. Suppose we suspect a common structural break in the mean at date τ. Define the post-break dummy

$$D_t \;=\; \mathbf{1}\{t \geq \tau\}.$$

We estimate the pooled regression

$$r_{i,t} \;=\; \alpha_i \;+\; \beta\, D_t \;+\; u_{i,t},$$

where α_i are factor fixed effects capturing pre-break mean levels, β is the common mean shift after τ, and $u_{i,t}$ is an error term.

- Null hypothesis (no break):

  $$H_0 : \;\beta = 0.$$

- Alternative (mean shifts after τ):

  $$H_1 : \;\beta \neq 0,$$

  or, in a one-sided form relevant for Sharpe ratio declines,

  $$H_1 : \;\beta < 0.$$

The test statistic is the usual t–statistic on $\hat{\beta}$, with standard errors clustered by time to allow for cross-sectional dependence.

Table ?? presents the change point estimates from our test. Consistent with the prediction in Pénasse (2022), the decline in expected returns generates a short-lived spike in realized returns before the structural break materializes in 2002. This pattern aligns closely with what we document in the data.

Table 1: Pooled Common Break Test Results

	1995 (Pre)	2002 (Post)
CTF Non-Sparse SDFs	0.399***	-0.520***
	(3.72)	(-9.01)
CTF SDFs	0.024	-0.627***
	(0.30)	(-14.47)
JKP Factors PCs	0.005*	-0.003**
	(1.35)	(-2.31)
JKP Factors USA	0.002**	-0.001***
	(1.93)	(-4.11)
JKP Factors USA VM	-0.367	-0.805***
	(-3.23)	(-11.93)
JKP Themes Developed	0.001	0.000
	(0.78)	(0.11)
JKP Themes USA	0.002*	-0.001***
	(1.32)	(-2.52)
JKP Themes USA SDFs	0.085***	-0.004*
	(3.19)	(-1.58)

Notes: Entries report the coefficient on the period dummy; t-statistics in parentheses. Stars denote significance based on the reported one-sided p-values: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Baseline period is dates < 1995.

3 A Baysian IPCA

From Kelly et al. (2019) we know that IPCA is highly effective at pricing factor returns, both in the time series and in the cross-section. Consequently, if the risk premia—the means of the factors—have declined, such a decline must be reflected either through changes in the magnitude of the betas or through the IPCA-implied stochastic discount factor (SDF). Given our extensive prior empirical evidence, it is reasonable to attribute the decline to the latter, namely, a shift in the SDF. Building on this robust empirical observation that IPCA successfully prices the cross-section, and drawing inspiration from ? and Marrow and Nagel (2024), we extend IPCA into a Bayesian framework. Specifically, we endow a hypothetical agent with IPCA and a prior on the SDF means in order to estimate the *real-time* expected returns on the factors.

3.1 Baseline IPCA model

Following Kelly et al. (2019), excess realized returns satisfy

$$r_{i,t+1} \;=\; \alpha_{i,t} + \tilde{\beta}'_{i,t} f_{t+1} + \tilde{\varepsilon}_{i,t+1}, \tag{1}$$

where $r_{i,t+1}$ is the excess return of asset i from t to $t+1$, $f_{t+1} \in \mathbb{R}^K$ are K latent factors, and $\tilde{\beta}_{i,t}$ are the time-t loadings.

Kelly et al. (2019) show (and we confirm) that the restricted model holds, so that $\alpha_{i,t} = 0$, and that loadings depend linearly on observable characteristics:

$$\tilde{\beta}_{i,t} \;=\; z'_{i,t}\Gamma + \nu_{i,t}, \tag{2}$$

with $z_{i,t}$ the $L \times 1$ vector of characteristics for asset i and Γ an $L \times K$ parameter matrix mapping characteristics into factor loadings.

We can then aggregate the latent factors into a single stochastic discount factor (SDF)

$$m_{t+1} \;=\; w'_f f_{t+1},$$

where w_f are the optimal factor weights (e.g. Hansen–Jagannathan tangency weights). Substituting yields the single-factor representation

$$r_{i,t+1} \;=\; \beta_{i,t}\, m_{t+1} + \varepsilon_{i,t+1}, \tag{3}$$

where the effective univariate beta is

$$\beta_{i,t} \;=\; \frac{\mathrm{Cov}(r_{i,t+1}, m_{t+1})}{\mathrm{Var}(m_{t+1})} \;=\; \frac{w'_f \Sigma_f\, \tilde{\beta}_{i,t}}{w'_f \Sigma_f w_f}. \tag{4}$$

Here $\Sigma_f = \mathrm{Var}(f_{t+1})$. Because $\tilde{\beta}_{i,t}$ is linear in $z_{i,t}$, the composite beta $\beta_{i,t}$ remains a linear index in characteristics as well. The residual $\varepsilon_{i,t+1}$ in the single-factor regression is not identical to the multi-factor residual $\tilde{\varepsilon}_{i,t+1}$. Instead,

$$\varepsilon_{i,t+1} \;=\; \tilde{\varepsilon}_{i,t+1} + \tilde{\beta}'_{i,t} f_{t+1} - \beta_{i,t} m_{t+1},$$

so while both residuals are orthogonal to their respective right-hand-side factors, they differ in distribution. The equivalence lies in mean pricing: $\mathbb{E}[r_{i,t+1}]$ is the same under the K-factor and single-factor representations, hence the pricing errors (alphas) coincide.

Remark: This logic extends to any high-powered SDF. Conditional on the agent knowing the SDF, they can back out dynamic single-factor betas that (by construction) inherit the ability to price the same cross-section as the original multi-factor representation.

Agents expected return is given by the conditional time t expectation

$$E_t[r_{i,t+1}] = \beta_{i,t} E_t[m_{t+1}]. \tag{5}$$

where

$$E_t[m_{t+1}] = \mu_t + u_{t+1} \tag{6}$$

3.2 Modeling Realized returns

There are two scenarios that may happen.

- Agent does not understand alpha decay then

$$r_{i,t+1} = \beta_{i,t} E_t[m_{t+1}] + \epsilon_{i,t+1} \tag{7}$$

- Agent understands realized returns are driven by alpha decay

$$r_{i,t+1} = \beta_{i,t} \left(E_t[m_{t+1}] - \delta(E_t[m_{t+1}] - E_{t-1}[m_t]) \right) + \epsilon_{i,t+1} \tag{8}$$

  δ could be measured with some duration measure at the SDF level (linear aggregation of stock level durations) [interesting to check out factor level duraiton as well and see if it is related to the pre break increase in returns.]

Agent then has a decaying prior for the mean of the SDF

$$m_{t+1} = \rho m_t + (1-\rho)\bar{m} + u_{t+1} \tag{9}$$

0 where $\bar{m}$ is the long run risk premium level.

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